

CogSci Experiment: Effectiveness of Learning Modalities

Peter Trinh

CS 6795 Intro to Cognitive Science

Georgia Institute of Technology

pтрinh32@gatech.edu

Abstract—This experiment researches how different instructional modalities of text only, audio only, and interactive multimedia learning can affect a learner’s comprehension and memory retention. The experiment used basic geometry as the lessons that participants learned within the experiment. Many theories such as the cognitive theory of multimedia and the dual coding theory supports the claims that the effective of learning modalities varies based on the type. The experiment aimed to determine whether which instruction type provides the best measurable benefits. Ten participants completed the experiment in which they were given three short lessons on area of 2D shapes, circumference and perimeter of 2D shapes, and volume of 3D shapes, where each topic was paired with a different modality. The participants would then complete a quiz immediately after the lesson and another quiz and survey one to two days afterword on the material learned. They results showed a clear and consistent pattern across all three types where text based instruction scored the lowest, audio only instruction had a slightly higher score, and the interactive modality scoring the best. Survey responses also indicated that interactive lessons were perceived as easiest, most engaging, and most confidence building. These findings suggest that interactive learning techniques promote a stronger understanding and memory of the material over text or audio alone, which can have great beneficial implications within the current education system.

I. INTRODUCTION

A. Research Question

This experiment focused on determining the research question: How does the modality of instruction, specifically text only, audio only , and interactive learning, influence learning comprehension and memory recall? The research aimed to identify the best learning modality by comparing the three different learning methods based on cognitive load, accuracy of memorization, and depth of comprehension.

B. Learning in Cognitive Science

Learning and education have always been core aspects of cognitive science since it provides a process in which the mind is able to represent, manipulate, store, and retrieve information. Our minds are constantly active in learning new information. Biology explains that the mechanisms behind our memories are chemical and biological processes; however, cognitive science gives us a different answer in which our memories revolve around the processes of understanding and reasoning. In addition, the process of learning can differ depending on the individual and the modality used for learning. Some learners excel with visual input, some learn better with auditory input,

and others learn best through interactive or hands on work. According to the Cognitive Theory of Multimedia Learning, people learn more deeply when material is presented through coordinated verbal and visual channels rather than one modality alone [1]. Based on this theory, we can see that depending on the instructional modality an individual’s mental processing can be improved or worsened. The Dual Coding Theory also has a similar stance in that information encoded through both verbal and non verbal representations leads to stronger recall because it creates multiple retrieval pathways [2]. The Cognitive Load Theory also expands on this by explaining that there are specialized limits of working memory based depending on cognitive load and that instructional materials can affect the cognitive load of the mind unnecessarily if not designed properly [3]. These theories all suggest that different types of environments may promote learning depending on the individual and that multimodal environments can engage cognitive systems better. Understanding how modality shapes learning has become significantly more important and practical in helping the education system.

C. Cognitive Theories Related to Learning Modalities

The experiment of determining the effectiveness of different learning modalities stems from the concepts of cognitive science support the idea that the learning process differs depending on the modality. Each theory gives insight into another aspect that supports the overall idea.

1) *Cognitive Theory of Multimedia Learning*: This theory from Mayer posits that learning is improved when verbal and visual information are presented in integrated ways that complement one another [1]. This can explain which type of modality would be best for learning.

2) *Dual Coding Theory*: The Dual Coding Theory explains that encoding information in both verbal and non verbal systems enhances the retrieval strength of the information [2]. This means that learning using visual aspects along with text or narration will greatly improve memorization of the data.

3) *Cognitive Load Theory*: This theory also focuses more on the memorization aspect of learning. The Cognitive Load Theory explains that the design of the information being learned affects the cognitive load of the mind itself. Poor designs overload the working memory, while multimedia is able to reduce strain and support cognitive effort [3].

4) *Interactive Multimodal Learning*: Moreno and Mayer's studies show that the theory that multimodal learning, where the individual has a hands on learning experience can increase engagement, transfer, and retention of information within the mind [4]. This supports the inclination to test the effectiveness of this interactive learning against other modalities.

D. Importance

The education system that we currently have within the United States focus traditionally on text based learning coupled with auditory lesson such as lectures and verbal explanations. A prime example would be this exact class in which the class itself comprises of mainly reading articles and textbooks, writing assignments, and watching lecture videos. However, the access to digital tools have now given us access to a possibly much more multimodal learning experience. The question is whether or not these teaching methods are truly viable, and whether or not students are learning better using prior techniques. Many research indicates that interactive learning can enhance engagement, participation, and comprehension of the materials [4]. While other studies about the modalities show that presenting information through complementary channels reduce cognitive load and increases comprehension, particularly for complex or unfamiliar material [5]. However, despite these findings, many education systems still rely heavily on singular modal teaching techniques. Throughout the education system there are many students who can be labeled as bad students because they struggle with traditional formats of learning, when in reality their difficulty stems from a mismatch of modality and cognitive processing. These misunderstood students are not be given the best possible environment to shine and the teaching methods are hindering their cognitive processing instead of helping. Research on inclusive instructional design emphasizes that multimodal approaches support a broader range of learners, including those with diverse cognitive profiles, language backgrounds, and learning disabilities [6]. These incorrect assessments of students and learning creates an environment in which ineffective learning can stump gifted students while widening the achievement gaps of others. We can greatly reduce these inequities within the system by analyzing the modalities of each type and learn how to most effectively use each type with each unique individual student. By examining how modality affects comprehension and memory in a controlled experiment, we can better create a more inclusive and cognitively effective learning environment for all students. There is an abundance of research and supporting theories that suggest that instructional modalities need to advance and adapt to the student, and this experiment aims to investigate how text, audio, and interactive formats influence comprehension and memory in a controlled setting to best improve the educational system.

II. EXPERIMENT DESIGN

A. Procedure

To examine how instructional modality influences learning comprehension and memory, this study controlled cognitive

experiment on learning using a controlled environment in which three different instructional types were used. These three types of modalities included text only learning, audio only learning, and interactive multimedia learning. The topic that was chosen to be used as the concept that participants will be learning was foundational geometry concepts. For each different type of modality, three distinct but comparable lessons within geometry were taught: area of 2D shapes, circumference and perimeter of shapes, and volumes of 3D shapes. This study used a repeated measure design so that every participant experienced all three learning modalities. This allowed the study to minimize individual differences in background knowledge and learning ability among the students. This ensured that the performances of the participants can be attributed primarily to the modalities instead of individual characteristics or math levels.

The experiment was conducted in two sessions. The first session required all participants to learn the materials of the concept. Immediately after the completion of the lesson, they would take comprehension quizzes involving the subjects they just learned. This would be repeated for all three types of modalities. The second session would take place 24 to 48 hours after the first session. This is the delayed recall quiz to assess memory retention. In addition, the participants also filled out a survey to explain their subjective feelings on the modalities and any feedback they would like to give.

B. Participants

The experiment had a total of 10 participants engaged in the study. All ten individuals were aged between 18 and 30 with at least a high school degree. The demographics of the sample of participants were split evenly between men and women, while the age range was not restricted. The requirement for eligibility of the research would be to have at least a high school level of education and the completion of a geometry focused class. They also needed to be fluent in English, since the instructional materials and audio narration were provided in English. Students with any level of visual or auditory impairments were excluded since the learning cognition may be affected in unknown ways. The recruitment of the participants were recruited through internal audits of the individual and personal recommendations since the experiment required an extensive in person learning and quizzing sessions. The participants needed to be able to meet up in person while having at least 2 hours available for 2 different days. The severely restricted the number of possible participants and required me to reach out to volunteers in person to find eligible people. While the number of participants representing the sample of the population is small, the number of participants can be argued to be enough since each participant contributes data for all three conditions, improving sensitivity to modality differences. This also compensates for the small N through reduced variance. However, a larger study with more participants would be recommended for stronger results and broader generalization.

C. Hypothesis

My hypothesis for the research experiment focuses on success of the interactive multimedia instruction. I hypothesized that the interactive modality would lead the highest levels of comprehension and memory retention, followed by audio only instruction, and text only instruction with the lowest score. Following the Cognitive Theory of Multimedia and the Dual Coding Theory, I based my predictions on the overwhelming supporting data that suggest that a mix of verbal and visual information learning modality would yield the best learning results.

1) Independent Variables:

- Text only modality
- Audio only modality
- Interactive multimedia modality

2) Dependent Variables:

- Immediate comprehension score
 - Accuracy on 5 question quizzes following each learning instruction
 - Measures initial understanding and ability to apply formulas
- Delayed recall score
 - Accuracy on 3 question quizzes 1-2 days later after learning instruction
 - Measures retention and memory development
- Subjective difficulty rating
 - Participant self report on perceived cognitive effort
- Engagement rating
 - Participant self report on how engaging each modality felt
- Confidence rating
 - Participant self report on how well they believed they understood the material

The immediate and delayed quiz scores focused on quantitative analysis to determine results. The different types of instructions will be compared through statistical tests that look for differences in average performance to draw assumptions from. Subjective data provided supportive qualitative evidence about the participants' perceived effects of each modality and the effectiveness of each type. It also helped determine the cognitive load of each instruction. The quantitative and qualitative data collected comes together to provide an educated conclusion on the understandings of how modality affects both the cognitive aspects of learning.

D. Process of Experiment

The process of the experiment followed a very strict time schedule to be control the exposure to each learning modality. Upon arrival, participants were informed of the process of the experiment and the topics that they would be learning. The order of the modalities taught were randomly selected so that the order in learning would not affect the results. In addition, each topic was then paired with a different modality

for each participant so that the topic itself could not influence the modality comparison.

For each learning session, a 15 minute block of time is cut out specifically to control the learning speed. In the text only modality, participants read a written lesson containing definitions, formulas, worked examples, and guided practice. The audio only lesson contained identical content that is narrated out loud to the participant while they are blindfolded so that there is no visual input. In the interactive lesson, participants viewed text explanation along with audio narration and completed a step by step practice problem. Once each learning session is complete, participants would immediately complete a 5 question quiz assessing their comprehension and problem solving with the topic they just learned. The quizzes are based on the topic they had just completed without any further concepts outside of the learning session. The questions assessed conceptual understanding, procedural ability, and formula application.

After completing all three learning modalities, the participants would come back after 1-2 days for the delayed recall session. The follow up would require the participants to complete a 3 question quiz of each topic learned from the three modalities again. The questions would follow along the same thought process and concept as the previous quizzes. This measured how much information had been retained across time from each modality. The delayed recall is an important measure in cognitive science because it reflects long term memory development, which is one of the aspects the experiment is trying to determine. In addition to the quiz the participants are also asked to complete a survey. At the beginning of the survey, every participant is asked about more generalized questions to get short understanding of the demographics of the participants, such as age, gender, major, experience with geometry, and math comfort levels. This information helped to provide context for interpreting the results. The rest of the survey revolves around rating the perceived difficulty, engagement, and confidence the participant has in their understanding of the topic. These subjective measures provide additional insight into how they experienced each modality cognitively and how they feel about each type emotionally.

E. CITI Training

The Citi Training for Group 2: Social/Behavioral Research Investigators and Key Personnel for Human Subjects Research was completed in a previous class on September 10, 2023 with an expiration on September 10, 2026. The certificate can be seen within the appendix.

III. RESULTS

The dataset below is collected through the process of a counterbalanced processing so each modality is rotated with each topic, with a total of 6 possible rotations. Each participant will be identified through number assignment, P01 - P10. This ensures that every topic appears under every modality and that all modalities occur in all positions.

A. Rotation Set Used

- 1) Area=Text, Circ=Audio, Volume=Interactive
- 2) Area=Audio, Circ=Interactive, Volume=Text
- 3) Area=Interactive, Circ=Text, Volume=Audio
- 4) Area=Text, Circ=Interactive, Volume=Audio
- 5) Area=Audio, Circ=Text, Volume=Interactive
- 6) Area=Interactive, Circ=Audio, Volume=Text

To prevent any single topic from influencing the results and to ensure that the differences in the data could be attributed to modality rather than content, the experiment used a counterbalanced repeated measure design. Every participant ultimately completed all three types of modalities with three lessons on area, circumference, and volume, but the topics and modalities were rotated and paired differently. A participant may learn area with text while another participant would learn it through audio or interactive. This aims to spread the modalities learned by with each topic so that they are evenly spread across all 10 individuals as best as possible. This helps reduce order effects on the data collected and gives a better comparison for the modalities.

B. Participant Quiz Scores (0-5)

- 1) P01 – Rotation 1:
 - Area = Text → 4
 - Circ = Audio → 5
 - Volume = Interactive → 5
- 2) P02 – Rotation 2:
 - Area = Audio → 4
 - Circ = Interactive → 5
 - Volume = Text → 4
- 3) P03 – Rotation 3:
 - Area = Interactive → 5
 - Circ = Text → 4
 - Volume = Audio → 4
- 4) P04 – Rotation 4:
 - Area = Text → 4
 - Circ = Interactive → 5
 - Volume = Audio → 4
- 5) P05 – Rotation 5:
 - Area = Audio → 5
 - Circ = Text → 4
 - Volume = Interactive → 5
- 6) P06 – Rotation 6:
 - Area = Interactive → 5
 - Circ = Audio → 4
 - Volume = Text → 3
- 7) P07 – Rotation 1:
 - Area = Text → 4
 - Circ = Audio → 5
 - Volume = Interactive → 5
- 8) P08 – Rotation 2:
 - Area = Audio → 5
 - Circ = Interactive → 5
 - Volume = Text → 4

9) P09 – Rotation 3:

- Area = Interactive → 5
- Circ = Text → 4
- Volume = Audio → 4

10) P10 – Rotation 4:

- Area = Text → 4
- Circ = Interactive → 5
- Volume = Audio → 4

C. Delayed Quiz Scores (0-5)

ID	Area	Circ	Volume
P01	2	3	3
P02	3	3	2
P03	3	2	2
P04	2	3	3
P05	3	2	3
P06	3	2	1
P07	2	3	3
P08	3	3	2
P09	3	2	2
P10	2	3	3

TABLE I

THIS TABLE FOLLOWS THE ROTATION OF EACH PARTICIPANT

D. Quiz Means

1) Participant Quiz Means (0-5):

- Text = 3.9
 - Standard Deviation = 0.32
- Audio = 4.4
 - Standard Deviation = 0.52
- Interactive = 5.0
 - Standard Deviation = 0.00

2) Delayed Quiz Means (0-3):

- Text = 1.9
 - Standard Deviation = 0.32
- Audio = 2.7
 - Standard Deviation = 0.48
- Interactive = 3.0
 - Standard Deviation = 0.00

E. Modality Survey Ratings

- Text
 - Difficulty = 2.8
 - Engagement = 2.4
 - Confidence = 3.4
- Audio
 - Difficulty = 2.1
 - Engagement = 3.9
 - Confidence = 4.4
- Interactive
 - Difficulty = 1.4
 - Engagement = 4.9
 - Confidence = 5.0

For these ratings, the lower the score for Difficulty the easier the modality was for learning.

F. Quiz Results

1) *Participant Quiz Results:* The quiz scores for the initial quizzes can be seen as decently high across all modalities. This is expected since the students had just learned the materials and the concepts involved basic geometry. Even then, there were still some deviation of the scores based on modality. The text only instructions had participants scoring with an average of 3.9 out of 5, which gives us a standard deviation of 0.32. Most people scored a 4 out of 5, with only one person scoring lower at a 3. This indicates that they were able to learn the material and apply the formula to the questions, just with simple problems with some mathematical errors. The audio version resulted in a higher average of 4.4 with a standard deviation of 0.52. The score was a higher level since everyone scored 4 out of 5 for this round. The strongest results came from the interactive instructions where every single participant got every question right with a average of 5. This produced a clear separation between modalities where text yielded the lowest mean, audio produced a higher, mean and interactive had the highest score.

2) *Delayed Quiz Results:* The delayed quiz results also had similar results as the participant quizzes. Even with fewer questions, the differences between the modalities became more apparent. the text only instructions gave an average score of 1.9 out of 3 with a standard deviation of 0.32. This indicates that some information from the text based lesson was retained but some could not remember all the formulas. The audio instructions had a slightly higher mean of 2.7 with a standard deviation of 0.48. Most participants were able to score higher than the previous instruction, which showed a more sustained memory development with this modality. The interactive version once again had a perfect score of 3 out of 3 on the delayed quiz showing the most retention from this type of learning. When comparing the immediate and delayed scores, all modalities showed some level of retention lost, except for the interactive type.

3) *Survey Rating and Subjective Response:* Looking at the survey ratings of the different modalities, we can see that the ratings align perfectly with the performance results of the learning types. Most participants rated text as the most difficult and least engaging, with only moderate confidence. Text based learning had a mean of 2.8 for difficulty, 2.4 for engagement, and 3.4 for confidence, where all levels were low. Audio lessons on the other hand had a much easier rating of 2.1 for difficulty, 3.9 for engagement, and 4.4 for confidence. Finally, we can see that interactive was by far the best with a easy rating of 1.4 in difficulty, 4.9 for engagement, and 5.0 for confidence. Based on the quiz results and survey rating, we get a pattern that shows that performance and experience of the participants for each modality goes from text, to audio, to interactive for effectiveness.

In addition, participants also wrote about their own personal feelings and thoughts about the modalities. Many wrote that they felt as though the text based lessons were harder to focus on and visualize since only reading over the lesson did not

stimulate them. They also specified that they had to read the lesson multiple times to actually retain and understand the material. For the audio lessons, there were similar complaints for the concepts. Participants were explaining that they had to relisten to the audio on more than one occasion to follow along with the concepts. They specified that the audio did not allow them to visualize the concepts at all. The interactive multimedia modality however did not have as many concerns, and some participants were able to specify that they were able to clearly understand the lesson and retain the material much easier. The interactive format was seen to be the most consistent in the subjective responses from the experiment with giving the most easy and effective way of learning.

IV. DISCUSSION

The findings from the study provide clear evidence that instructional modality plays a significant role in the comprehension and retention of materials learned by students. In the experiment, geometry was used as the concepts that were learned; however, this can be extended to other subjects as well. The tasks within the experiment were relatively simple and the differences in the learning conditions were strong enough to support the idea that the modality of learning influences the effectiveness. The results of the experiment leave a few key takeaways from that should be confirmed further.

The first implication would be the effectiveness of interactive multimedia instruction. The research clearly supports the difference in effectiveness of this modality. It consistently out scored text only and audio only instruction and the subjective responses from the participants also supported this idea. The interactive lessons used in this study required students to not only work with geometric shapes and formulas but also manipulate them through guided practice. This supports the Cognitive Theory of Multimedia Learning that claims that interactive multimedia can help reduce cognitive load and promote active processing. The findings also have implications for individual differences in learning.

Although the sample size was small, the overall subjective response of the participants have a general agreement that text only or audio only learning can be overwhelming and disengaged the learning process. They felt as though there were missing elements to the learning that can better help them understand the material. However, there were some students who did felt that audio or text only were still better learning experiences for them. This confirms the fact that modality preference can affect the effectiveness of the learning process, but overall the majority of people still prefer interactive materials. This supports the broader consensus in education that learning styles do not reliably predict actual performances, but may still influence the results.

Looking at results of the experiment, we can reinforce the fundamental understanding of learning in cognitive sciences. Interactive instruction supports the Dual Coding Theory since it presents information both verbally and visually. We can understand from this research that learners actively create

meaning through manipulation and feedback instead of just simply information absorption. The interactive multimedia modality required participants to engage and think about their actions and process. However, the text and audio only versions were a singularly focused on information processing. With such a shallow focus on the material the mind would have a slower encoding of the data and less retention overall.

The most important aspect of this research experiment is the possible real world applications of the results. The current education system still rely heavily on text based assignments and lectures. These formats are severely out of date and inefficient in learning, even though they are more efficient in distribution. We need to support the integration of more complex and interactive learning techniques that engage the students so that they can receive a more dynamic and comprehensive learning environment. The benefits of using interactive multimedia modality can stretch out further from just to education system to other programs as well, such as job trainings and tutoring schools. Interactive instructions has proven that it can lead to better learning experiences and effective learning environments, which the education system should adopt into its procedures.

V. CONCLUSION

The results of the experiment is clear and my hypothesis is confirmed to be correct. The instructional modality of interactive multimedia is determined to be the most effective learning instruction when compared with text only and audio only instructions. Across all participants, interactive instruction consistently produced the highest scores on both participant and delayed quizzes. This support is further confirmed with the participants' subjective ratings of each modality on easiest to learn from, engagement, and confidence. This all comes together to conclude that interactive is the most effective modality. Audio instruction follows in second with the next best results from the quizzes and the survey. Although text based learning is last, it does not fall behind audio only instructions by too much. The results of the experiment also strongly supports and reinforce the principles and theories of cognitive science in learning, such as the Dual Coding Theory and the Cognitive Load theory. The data collected from the learning modalities in the experiment suggest that there is strong practical applications that can be used in the education system from this research. However, further research would be required on a more extensive scale and a wide population sample. Although, future applications of this research could involve expanding the interactive components to include adaptive feedback or introducing digital technologies such as augmented reality into the teaching instructions. As new digital technologies are introduced further, the instructional methods that we use need to further adapt and evolve to best stimulate the most effective learning environment for students.

VI. LIMITATIONS

Although the results of the experiment is clear, there are a few limitations on the study that restrictions the overall

conclusions and may have affect the results. The main issue with the current experiment is that the sample size that was used is too small, with only 10 participants. This can severely limit the generalization of the findings of the research. A larger and more diverse pool of participants would help strengthen the statistics and confirmation of the data. Any future studies should spread out the participant pool and hopefully draw upon a wider age range and educational background as well. Another limitation of this study was the depth of the learning modalities available with the topic. Due to time constraints, the topic used for the learning modalities were deliberately simple for the educational background of the participants. Ideally, the experiment would have tested multiple subjects at varying levels of difficulty to get a wider range of data on multiple topics to best identify patterns in the modalities. Finally, the study relied on simple quizzes to measure learning, which is not the most in depth method to measure the conceptual understanding a person has on a subject. Future work should examine more complex comparisons of the participants in full application of the concepts to best measure whether the modalities have a strong enough difference in comprehension and memory retention. In summary, while the study provides evidence that interactive learning is most effective, further research with larger samples would best give a stronger understanding of the effectiveness of different learning modalities.

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APPENDIX A PARTICIPANT QUIZZES

A. Quiz A: Area

- 1) Find the area of a rectangle with length 8 cm and width 5 cm.
- 2) Find the area of a triangle with base 10 m and height 4 m.
- 3) Find the area of a circle with radius 6 cm.
- 4) A square has area 64 m². What is its side length?
- 5) A triangle has area 30 cm² and base 12 cm. What is its height?

B. Quiz B: Circumference and Perimeter

- 1) Find circumference of a circle with radius 7.
- 2) Find circumference of a circle with diameter 12.
- 3) A triangle has sides 5, 6, and 7. Find its perimeter.

- 4) A square has perimeter 48. What is its side length?
- 5) A circle has circumference 31.4. Estimate the radius.

C. Quiz C: Volume

- 1) Find the volume of a rectangular prism $5 \times 4 \times 3$.
- 2) Find the volume of a cube with side 6.
- 3) Find volume of a cylinder with radius 3 and height 10.
- 4) A rectangular prism has volume 120 and base 12×2 . Find height.
- 5) A cylinder has volume 150 and radius 2. Find height.

D. Delayed Quiz A

- 1) Find area of rectangle 12×7 .
- 2) Find area of triangle base 14, height 9.
- 3) Circle radius 5: find area.

E. Delayed Quiz B

- 1) Circle radius 9: find circumference.
- 2) Rectangle 6×11 : perimeter?
- 3) Triangle sides 9, 10, 11: perimeter?

F. Delayed Quiz C

- 1) Volume of prism $7 \times 5 \times 6$.
- 2) Cylinder radius 4, height 12: volume?
- 3) Cube side 9: volume?

APPENDIX B POST EXPERIMENT SURVEY

A. Section 1: Demographics and Background

- 1) What is your age?
 - 18–19
 - 20–21
 - 22–23
 - 24–25
 - 26 or older
- 2) What is your academic major? (Short answer)
- 3) How confident are you in your general math abilities?
 - 1 = Not confident at all
 - 2 = Slightly confident
 - 3 = Moderately confident
 - 4 = Very confident
 - 5 = Extremely confident
- 4) Before this study, how familiar were you with the geometry topics (area, circumference, volume)?
 - 1 = Not familiar
 - 2 = Slightly familiar
 - 3 = Moderately familiar
 - 4 = Very familiar
 - 5 = Extremely familiar

B. Section 2: Experience with Modalities

- 1) Which instructional modality felt the easiest to learn from?
 - Text-only
 - Audio-only
 - Interactive
 - All felt similar
 - None felt easy
- 2) Which instructional modality felt the most engaging?
 - Text-only
 - Audio-only
 - Interactive
 - All felt similar
 - None felt engaging
- 3) Rate how difficult each modality felt to use: (*1 = very easy, 5 = very difficult*)
 - Text-only: 1 2 3 4 5
 - Audio-only: 1 2 3 4 5
 - Interactive: 1 2 3 4 5
- 4) Rate how confident you felt in your understanding after each modality: (*1 = not confident, 5 = extremely confident*)
 - Text-only: 1 2 3 4 5
 - Audio-only: 1 2 3 4 5
 - Interactive: 1 2 3 4 5
- 5) Which modality do you believe helped you understand the geometry concepts the best?
 - Text-only
 - Audio-only
 - Interactive
 - They were all equal
 - I am not sure

C. Section 3: Feedback

- 1) In your own words, describe what helped or hindered your learning in the different modalities. (Short answer)

APPENDIX C CITI TRAINING CERTIFICATE



Completion Date 10-Sep-2023
Expiration Date 10-Sep-2026
Record ID 58272659

This is to certify that:

Peter Trinh

Has completed the following CITI Program course:

Not valid for renewal of
certification through CME.

Human Research
(Curriculum Group)
Group 2 Social / Behavioral Research Investigators and Key Personnel
(Course Learner Group)
1 - Basic Course
(Stage)

Under requirements set by:

Georgia Institute of Technology

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APPENDIX D
RESEARCH PLAN TASK LIST

Week	Task	Task Description	Estimated Time (Hours)	Complete? (Y/N)	Total Hours
3	1	Create the template task list.	0.25	Y	0.25
3	2	Choose research question	0.25	Y	0.25
3	3	Initial research question research	4	Y	5
4	4	Project Pitch Introduction	1.5	Y	2
4	5	Initial Research Design	1	Y	1
4	6	Citi Training	0.5	Y	0.5
5	7	Complete template task list	1	Y	1.5
5	8	Research on topic	4	Y	5
6	9	Participant Recruitment	1	Y	4
6	10	Procedure Development - text based learning	4.5	Y	6
6	11	Participant Experiment Control Individual	2	Y	2
6	12	Procedure Refinement	1	Y	3
7	13	Participant Recruitment	1	Y	3
7	14	Procedure Development - audio based learning	4.5	Y	5
7	15	Participant Experiment Control Individual	2	Y	2
7	16	Procedure Refinement	1	Y	3
8	17	Participant Recruitment	1	Y	6
8	18	Procedure Development - interactive learning	4.5	Y	11
8	19	Participant Experiment Control Individual	2	Y	3
8	20	Procedure Refinement	1	Y	4
9	21	Participant Experiment - Participants 1-3	6	Y	12
9	22	Data Collection	1	Y	2
10	23	Participant Experiment 4-7	6	Y	10
10	24	Data Collection	1	Y	2
11	25	Participant Experiment 8-10	6	Y	11
11	26	Data Collection	1	Y	2
12	27	Quantitative Data Analysis	2	Y	3
12	28	Qualitative Data Analysis	2	Y	2
12	29	Develop Visual Data Analytics	2	Y	4
12	30	Draft Final Report	5	Y	8
13	31	Quantitative Data Analysis	2	Y	4
13	32	Qualitative Data Analysis	2	Y	3
13	33	Develop Visual Data Analytics	2	Y	5
13	34	Continue Writing Final Report	5	Y	13
14	35	Complete writing Final Report	10	Y	16
14	36	Proofread Final Report	1	Y	4
15	37	Develop Powerpoint Presentation	5	N	
15	38	Write Presentation Script/Talking Points	1	N	
15	39	Practice Presentation	1	N	
15	40	Record Presentation	1	N	
			100		168.5